

Notes: Liao, Liu, and Wang (JFE, 2014)

China's Secondary Privatization

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1 What the paper is doing

This paper treats China’s 2005–2007 Split-Share Structure Reform (SSSR) as a large-scale policy shock that *granted legal trading rights to previously non-tradable state-owned shares*. The reform is interpreted as “opening the gate” to *secondary privatization*: even if the state does not immediately sell control, the *expectation* that it *can* sell and that share value matters may reshape incentives inside state-owned enterprises (SOEs).

The core idea is an incentive story:

- **Before the reform:** the state’s shares are largely illiquid (non-tradable). A controlling state shareholder can extract private benefits or pursue non-value objectives with weaker market discipline.
- **After the reform:** non-tradable shares become tradable after a lockup. This makes the state-share value “monetizable” and connects the government’s and managers’ payoffs more tightly to market valuation.
- **Prediction:** even *before* actual divestment, the *expectation of future privatization* can raise incentives to boost performance and share value.

Empirically, the paper asks three linked questions:

1. **Is there a privatization effect?** Did SOEs improve more than comparable non-SOEs after the reform?
2. **Where does it come from?** Are SOE improvements linked to government agents’ supportive actions (e.g., asset injections, group listings, financing opportunities)?
3. **Did the reform’s market mechanism matter?** Do negotiated terms (consideration and approval rates) predict subsequent performance in a way consistent with information discovery and incentive communication?

2 Institutional background: what is the Split-Share Structure Reform?

China’s initial share issue privatization (SIP) created a dual-share structure:

- **Tradable shares:** held by public investors, priced in the market.
- **Non-tradable shares:** mainly state-owned (and some legal-person shares), not freely tradable in the secondary market.

This “split-share” structure generated severe frictions: misaligned incentives, corporate governance problems, speculative trading, and (crucially) a large “overhang” of potentially sellable state shares.

The SSSR (initiated in 2005) converted non-tradable shares into legitimate tradable shares *via a negotiated mechanism*:

1. Non-tradable shareholders propose a reform plan, including **consideration** paid to tradable shareholders (cash, shares, options, warrants, or promises such as future dividends / asset injections).
2. Tradable shareholders negotiate and then vote; the plan requires a supermajority approval among participating tradable shareholders.
3. After approval, non-tradable shares face a **lockup period** (at least 12 months), and subsequent sales are constrained (e.g., volume limits and disclosure rules).

This negotiation/vote feature is a key “market mechanism” in the paper: it creates observable prices/terms (consideration, approval rates) that can embed information and expectations about future performance.

3 Hypotheses

The paper formalizes three hypotheses:

- H1:** After the reform, **SOEs improve performance more than non-SOEs** (a positive “privatization effect”).
- H2:** **Privatization-led improvements** to SOE performance are **positively correlated with government agents’ supportive activities** (e.g., group listing, asset injection, financing opportunities).
- H3:** Privatization-led improvements are **negatively correlated with consideration** and **positively correlated with approval rates**, consistent with the reform’s **market mechanism** facilitating information discovery and incentive communication.

4 Data, sample, and measurement choices

4.1 Sample construction and groups

The main sample consists of **1,032 listed firms** that completed the reform during **2005–2007**. Firms are classified as:

- **SOE** if the *ultimate controlling party* is the state; otherwise **non-SOE**.
- **State-ownership** is measured as the ratio of state-owned shares to total shares outstanding.
- The paper also partitions firms into groups by state-ownership intensity: **P** (zero state ownership), **SL** (low), **SM** (medium), and **SH** (high).

4.2 Why medians and quantile regressions?

The paper emphasizes that the key variables (revenues, profits, returns) are *highly skewed and leptokurtic*. This motivates:

- Using **medians** rather than means in before/after comparisons;
- Using **Wilcoxon signed-rank** and **Wilcoxon Z-tests** for nonparametric inference;
- Estimating **quantile regressions** (Koenker–Bassett) to study heterogeneous effects along the outcome distribution.

4.3 Performance variables (and accounting constraints)

Because China revised GAAP around 2007, some accounting ratios are not comparable before/after (e.g., ROA, ROE, debt-to-asset, current ratio). The paper therefore focuses on:

- **Output:** CPI-adjusted operating revenue and operating profit.
- **Employment:** number of employees.
- **Investment:** capital expenditure normalized by operating revenue.
- **Productivity proxies:** operating revenue per employee; operating profit per employee.
- **Operating efficiency:** accounts receivable turnover, asset turnover, expense-to-sales ratio.
- **Insolvency risk:** cash-to-total-liabilities ratio (chosen for comparability).
- **Corporate governance / tunneling proxies:** related-party transactions, related-party transactions with controlling shareholders, and lending to controlling shareholders (plus intensity measures among firms engaged).

5 Models and empirical design

5.1 Benchmark-adjusted stock returns (Fama–French)

To study post-reform stock performance net of standard risk premia, the paper uses the Fama–French three-factor model. For firm i :

$$\hat{r}_i = \beta_i^M(r_M - r_f) + \beta_i^{SMB}r_{SMB} + \beta_i^{HML}r_{HML} + r_f. \quad (1)$$

The corresponding **risk-adjusted return** is $r_i^{adj} = r_i - \hat{r}_i$, and the paper constructs multi-year cumulative adjusted returns after the reform.

5.2 Differences-in-differences style “privatization-led” improvement

The paper measures the privatization-led improvement of an SOE by subtracting the median change among matched non-SOEs:

$$IMPPRIV_i \equiv \Delta Perform_i^{SOE} - \Delta Perform_{b(i)}^{BK}, \quad (1)$$

where $b(i)$ is the benchmark portfolio matched on **size** $\times$ **industry** (and, for robustness, **size** $\times$ **market-to-book**). The use of matched non-SOEs is meant to net out common macro shocks and non-privatization components of the reform.

5.3 H1: Does state ownership predict post-reform performance changes?

To test whether SOEs (and, more generally, firms with higher state ownership) improve more after the reform, the paper estimates:

$$\Delta Perform_i = \alpha + \beta^{StateOwn} StateOwn_i + \sum_{j=1}^N \beta_j^{Control} Control_{i,j} + \varepsilon_i. \quad (2)$$

Key prediction (H1): $\beta^{StateOwn} > 0$.

Quantile regression form. At quantile $\tau \in (0, 1)$, the conditional quantile is modeled as

$$Q_\tau(\Delta Perform_i | X_i) = X_i' \beta(\tau),$$

estimated by minimizing the Koenker–Bassett objective

$$\min_{\beta} \sum_i \rho_\tau(\Delta Perform_i - X_i' \beta), \quad \rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\}).$$

This is why Tables 6–7 and Figs. 1–2 often report results at the 25%, 50%, and 75% quantiles.

5.4 H2: Is the privatization effect stronger when government agents “support” SOEs?

Using the SOE subsample, the paper relates privatization-led improvements to proxies for government agents’ supportive activities:

$$\begin{aligned} IMPPRIV_i = & \alpha + \beta^{GroupList} GroupList_i + \beta^{AssetInject} AssetInject_i + \beta^{FundRaise} FundRaise_i \\ & + \sum_{j=1}^N \beta_j^{Control} Control_{i,j} + \varepsilon_i. \end{aligned} \quad (3)$$

Key prediction (H2): $\beta^{GroupList}, \beta^{AssetInject}, \beta^{FundRaise} > 0$. To avoid multicollinearity, the paper includes these proxies *one at a time* in Table 8.

5.5 H3: Does the market mechanism reveal information and incentives?

The reform plan embeds two observable negotiation outcomes:

- *Consider_i*: consideration paid to tradable shareholders;
- *Approval_i*: public-investor approval rate for the plan.

The paper tests:

$$IMPPRIV_i = \alpha + \beta^{Consider} Consider_i + \beta^{Approval} Approval_i + \sum_{j=1}^N \beta_j^{Control} Control_{i,j} + \varepsilon_i. \quad (4)$$

Key prediction (H3): $\beta^{Consider} < 0$ and $\beta^{Approval} > 0$.

6 Results: interpreting each table and figure

6.1 Table 1: Summary statistics

What it shows. Medians of key variables for the full sample (1,032 firms), split by (i) ultimate control (399 non-SOEs vs 633 SOEs) and (ii) state-ownership groups (P, SL, SM, SH).

What we take from it.

- **Scale:** SOEs are substantially larger (assets, revenue, employment, EBIT) than non-SOEs.
- **Ownership structure:** non-tradable shares are a large fraction of total shares (around 60% in medians), consistent with a big pre-reform “overhang.”
- **Incentives:** managerial shareholding is tiny in medians, suggesting weak direct equity incentives at the management level—which makes the reform’s incentive channel (via state-share value) more central.
- **State ownership gradients:** moving from P to SH tends to coincide with larger size and, in some profitability measures, higher medians, but the table is descriptive rather than causal.

6.2 Table 2: Output, employment, and productivity before vs after

Design. The table reports median changes from three years pre-reform to three years post-reform. It also shows “increase odds” (number of firms with increases vs decreases) and differences across groups.

Key patterns.

- **Output rises strongly.** Median operating revenue increases by about 73% in the full sample. SOEs rise more (84%) than non-SOEs (57%), with a significant difference (SOE–non-SOE $\approx$ 27 percentage points).

- **Profits also rise, but the SOE–non-SOE gap is smaller.** Median operating profit increases around 45% (full sample). The SOE–non-SOE difference is small and statistically weak in the median comparison.
- **Scale expansion.** Total assets and employment increase, and the increases are stronger for SOEs and for higher state-ownership groups ($P \rightarrow SH$).
- **Productivity proxies improve, but are not uniquely “SOE-driven.”** Operating revenue per employee increases around 41% across groups, with little SOE–non-SOE difference in median change.
- **Investment (capex normalized by revenue) declines in medians.** This is consistent with either (i) revenue growing faster than capex, (ii) restraint in expansion during the post-2007 crisis years, or (iii) composition changes in investment measurement.

Interpretation. The reform coincides with a broad improvement in output and employment, and the cross-sectional gradient in state ownership is consistent with H1’s incentive story (larger state stake $\Rightarrow$ more incentive to raise state-share value). At the same time, productivity improvements are broadly shared, foreshadowing the later result that operating efficiency does not change differentially for SOEs.

Table 1
Summary statistics

This table reports the summary statistics of the 1,012 sample firms that completed the light share structure reform during 2005–2007. Medians of variables are reported. A firm is labeled as SOE (non-SOE) if its ultimate controlling party is (not) the state. Groups P, S*, and S** represent the subsamples of firms with zero, low, medium, and high levels of state-ownership measured by the ratio of number of state-owned shares to total shares outstanding. Net margin rate is calculated as net income divided by revenue. Accounts receivable turnover, asset turnover, and inventory turnover are calculated using operating revenue divided by average accounts receivable, total assets, and inventory, respectively. Capital expenditure is calculated as change in gross properties, plant, and equipment plus change in intangible assets. EBIT represents earnings before interest and tax. On average, SOEs exchanged into 10087.04 during the sample period.

Variable	Full sample	By ultimate control		By state-ownership			
		Non-SOE	SOE	P	S*	S**	S** - P
Sample	Number of firms	5312	399	631	146	295	295
	Non-volatile share-to-total share ratio	01.85	00.35	02.25	00.00	38.03	55.98
Size	Total assets (RMB in millions)	1727.9	1204.5	2268.6	1298.5	2204.9	2203.44
	Revenue (RMB in millions)	186.41	193.31	123.60	92.83	69.95	135.39
	Number of employees	1408.54	1277.93	2288.69	1295.95	1455.99	1071.60
Profitability	EBIT (RMB in millions)	63.74	46.73	63.75	49.37	43.35	68.09
	Net margin rate (%)	3.46	3.66	3.30	3.24	3.22	4.72
	ROA (%)	2.16	2.48	2.30	1.80	1.82	2.22
	ROE (%)	4.95	4.25	5.30	4.55	3.39	4.97
Capital structure	Debt-to-asset ratio	0.52	0.55	0.51	0.54	0.54	0.48
	Current ratio	1.91	1.84	1.97	1.93	1.97	1.99
Growth	Market-to-book ratio	1.72	1.76	1.70	1.68	1.79	1.64
	Capital expenditure (RMB in millions)	61.20	31.07	87.98	42.59	40.88	79.23
Productive efficiency	Accounts receivable turnover	3.31	3.33	3.07	3.29	4.09	5.49
	Asset turnover	0.16	0.18	0.20	0.15	0.16	0.01
	Inventory turnover	3.90	3.36	4.32	3.13	3.77	3.69
Management incentive	Management shareholding (%)	0.04	0.05	0.04	0.09	0.07	0.01

Table 2
Post-reform changes in output, employment, and productivity

This table reports changes in firm output, profitability, employment, and productivity three years before and after the reform. The sample consists of 1,012 firms that completed the light share structure reform during 2005–2007. Wilcoxon signed-rank test is applied to examine the significance of changes in medians of variables. Wilcoxon Z-test is applied to examine the significance of difference in changes between groups. Proposition 2-test is used to test whether increase (decrease) in output is greater than that in the post-reform period. The p-values are reported in parentheses. The definitions of SOE and non-SOE (Groups P, S*, S**, and S** - P) are the same as in Table 1. A variable with “*” associated with percentage change indicates that the difference between post-reform and pre-reform values is statistically significant at the 10% level. Otherwise, the change is calculated as the third difference between post-reform and pre-reform values. Variables in bold are incommensurable before and after the reform of the Chinese Generally Accepted Accounting Principles effective in 2007, and reported only for reference. Superscripts “*”, “**”, and “***” denote the significance levels of 10%, 5%, and 1%, respectively.

Variable	Full sample	By ultimate control		By state-ownership						
		Increase ratio	Median change	Non-SOE	SOE	SOE - non-SOE	P	S*	S**	S** - P
Output & employment										
Operating revenue	0.41***	0.41***	0.37***	0.67***	0.30***	0.37***	0.67***	0.44***	0.07***	0.000
Operating profit	0.40***	0.40***	0.36***	0.67***	0.31***	0.36***	0.67***	0.44***	0.07***	0.000
Total assets	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.000
Capital expenditure	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.000
No. of employees	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.000
Capital expenditure (%)	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.000
Profitability										
Operating profit	0.41***	0.41***	0.37***	0.67***	0.30***	0.37***	0.67***	0.44***	0.07***	0.000
Operating profit per employee	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.13***	0.000
ROA (%)	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.000
ROE (%)	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.01***	0.000

Table 3

Post-reform changes in operating efficiency and insolvency risk

This table reports changes in firm operating efficiency and insolvency risk three years before and after the reform. The sample consists of 1,012 firms that completed the light share structure reform during 2005–2007. Wilcoxon signed-rank test is applied to examine the significance of changes in medians of variables. Wilcoxon Z-test is applied to examine the significance of difference in changes between groups. Proposition 2-test is used to test whether increase (decrease) in output is greater than that in the post-reform period. The p-values are reported in parentheses. The definitions of SOE and non-SOE (Groups P, S*, S**, and S** - P) are the same as in Table 1. The change is calculated as the level difference between post-reform and pre-reform values. Variables in bold are incommensurable before and after the reform of the Chinese Generally Accepted Accounting Principles effective in 2007, and reported only for reference. Superscripts “*”, “**”, and “***” denote the significance levels of 10%, 5%, and 1%, respectively.

Variable	Full sample	By ultimate control		By state-ownership					
	Increase ratio	Median change	Non-SOE	SOE	SOE - non-SOE	P	S*	S**	S** - P
Operating efficiency									
Accounts receivable turnover	0.0020***	4.51***	4.24***	4.64***	0.39	3.79***	5.20***	4.60***	0.82
Expense-to-sales ratio	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Asset turnover	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Inventory turnover	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Current ratio	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Insolvency risk									
Debt-to-asset ratio	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Debt-to-equity ratio	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
Current ratio	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001

6.3 Table 3: Operating efficiency and insolvency risk

Operating efficiency (Panel A).

- Accounts receivable turnover increases sharply (median change ≈ 4.5), indicating improved cash collection / revenue generation relative to receivables.
- Expense-to-sales ratio rises slightly (median $\approx +0.03$), which could reflect higher selling/financial expenses in the post-reform environment.
- Asset turnover increases (median $\approx +0.08$), indicating better revenue generation relative to assets.

Across-group comparison. The SOE–non-SOE difference is small and typically insignificant; state-ownership groups do not show a monotone pattern.

Insolvency risk (Panel B). The cash-to-total-liabilities ratio shows little systematic improvement; some group-level changes are mixed. Many standard leverage/liquidity ratios are flagged as accounting-incommensurable.

Interpretation. The reform period is associated with broad efficiency changes, but *not* a clear differential improvement in SOE efficiency. This matters because it pushes the privatization effect away from “pure efficiency” and toward “incentives for scale and performance”.

6.4 Table 4: Corporate governance and tunneling proxies

This table asks whether agency/tunneling behaviors decline after the reform.

Panel A (extensive margin): The fraction of firms engaging in:

- related-party transactions,
- related-party transactions with controlling shareholders, and
- lending to controlling shareholders,

declines after the reform. The most dramatic drop is **lending to controlling shareholders** (from about 42% pre to 17% post for the full sample).

Panel B (intensive margin among engaged firms): Median amounts involved (scaled by revenue) show mixed patterns. There is no consistent evidence that SOEs improve governance *more* than non-SOEs.

Interpretation. This is an important qualification: even if performance improves, the reform *does not robustly fix corporate governance* without deeper ownership changes. This aligns with the paper’s main narrative: incentives for performance can move quickly (expectations), but governance structures are stickier.

Table 4
Post-reform change in corporate governance.

This table reports changes in corporate governance measures before and after the reform. The sample consists of 1,032 firms that completed the Split-share Structure Reform during 2005–2007. Panel A reports the percentage of firms engaged in agency activities. Panel B reports the median ratios of the amount of money involved in agency activities to operating revenue for firms engaged in these activities. Wilcoxon Z-test is applied to examine the significance of the difference between groups in Panel B. The p-values are reported in parentheses. The definitions of groups P , S^L , S^M , and S^H are the same as in Table 1. Superscripts *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	Pre-reform						Post-reform					
	Full sample	P	S^L	S^M	S^H	$S^H - P$	Full sample	P	S^L	S^M	S^H	$S^H - P$
Panel A: Percentage of firms engaged in the agency activities												
Related-party transaction	43.1%	39.7%	36.9%	45.9%	49.2%	9.4%	35.7%	29.9%	32.5%	39.0%	38.3%	8.4%
Related-party trans. with controlling shareholders	29.7%	29.5%	22.7%	31.8%	34.9%	5.3%	24.0%	19.4%	20.0%	25.4%	28.8%	9.4%
Lending to controlling shareholders	42.3%	37.7%	40.0%	43.2%	46.1%	8.4%	16.6%	9.7%	11.5%	19.3%	22.4%	10.8%
Panel B: Median ratio of the amount of funds involved to operating revenue												
Related-party transaction (%)	6.92	6.90	7.60	7.76	5.84	−4.06*	7.16	8.34	7.17	5.47	7.74	−0.58 (0.660)
Related-party trans. with controlling shareholders (%)	5.81	5.59	6.34	5.66	5.84	0.25 (0.725)	7.36	10.70	5.40	6.48	9.15	−1.55 (0.412)
Lending to controlling shareholders (%)	1.11	1.01	1.43	0.64	1.15	0.14 (0.645)	0.11	0.49	0.07	0.14	0.10	−0.39 (0.436)

Table 5
The privatization effect.

This table reports the privatization effect measured by privatization-led post-reform changes in SOE operating revenue, operating profit, and stock return, respectively. The sample consists of 633 SOEs that completed the Split-share Structure Reform during 2005–2007. Unadjusted change in firm operating revenue (profit) is measured as the difference of the variable three years before and after the reform, normalized by pre-reform value. **Unadjusted stock return is the three-year cumulative Fama-French model-adjusted return after the reform.** In measuring the privatization effect, SOE operating revenue, operating profit, and stock return are further adjusted by the median operating revenue, operating profit, and stock return of a matching non-SOE benchmark portfolio by size–industry and size–market-to-book ratio, respectively. The p-values of Wilcoxon signed-rank tests on the significance of changes in the medians of variables are reported in parentheses. Superscripts *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	Unadjusted	Adj. by non-SOE size–industry portfolio	Adj. by non-SOE size–M/B ratio portfolio
Operating revenue	0.84*** (0.000)	0.06*** (0.000)	0.03*** (0.000)
Operating profit	0.50*** (0.000)	0.02*** (0.000)	0.000*** (0.000)
Stock return	1.09*** (0.000)	0.06 (0.431)	0.20** (0.019)

6.5 Table 5: The “privatization effect” (SOEs relative to matched non-SOEs)

Object. Median SOE changes are adjusted by matched non-SOE benchmark portfolios.

Results.

- Unadjusted SOE median changes are large: revenue $\approx 84\%$, profit $\approx 50\%$, and cumulative risk-adjusted stock return $\approx 109\%$.
- After adjusting by size $\times$ industry (or size $\times$ M/B) benchmarks, the **incremental** effect is smaller but still positive in many cases (e.g., revenue remains a few percentage points higher than benchmarks).

Interpretation. This provides the paper’s first “diff-in-diff” style evidence that SOEs improve more than comparable non-SOEs, consistent with H1.

6.6 Table 6 and Fig. 1: H1 via quantile regressions on revenue and profit changes

Table 6 (fundamental performance).

- **Operating revenue:** State-ownership has a positive and significant coefficient across quantiles (e.g., around 0.46 at the 25th percentile, 0.51 at the median, and 0.83 at the 75th percentile).
- **Operating profit:** The positive relationship is weaker at lower quantiles but becomes stronger at the upper quantile (state-ownership is most pronounced for high profit-growth firms).

Fig. 1 (heterogeneity). The figure traces the state-ownership coefficient across the full quantile spectrum:

- For revenue changes, the coefficient is positive across quantiles.
- For profit changes, the coefficient tends to be more positive in the right tail, highlighting that the “privatization effect” is stronger among firms with larger improvements.

Interpretation. Together, Table 6 and Fig. 1 provide the paper’s cleanest evidence for H1: *higher state ownership predicts stronger post-reform performance improvements*, consistent with an expectation-driven incentive channel.

Table 6
Post-reform changes in operational performance and state-ownership.

The table reports regression results of changes in firm operating revenue and operating profit on state-ownership for the full sample of 1032 firms that completed the light-share structure before during 2003–2007. We follow Gozgor and Isenhardt (2015) to apply quantile regression. The results for the 25th, 50th (median), and 75th quantiles are reported. OLS regression results are reported for reference. Data are estimated at the 1% level for OLS regressions. t-statistics are reported in parentheses. Significance: *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	Operating revenue				Operating profit			
	25th quantile	50th quantile	75th quantile	OLS	25th quantile	50th quantile	75th quantile	OLS
Intercept	-54.80***	-73.39***	21.68	24.64	-43.17**	-26.04	86.27	144.81***
State-ownership (S)	0.45**	0.53***	0.82**	0.51	0.18	0.37	0.74	0.27
Non-tradable-to-tradable ratio	0.75	0.75	0.51	1.15	0.64	0.51	0.24	0.30
Log of market cap.	0.46***	0.51***	0.51***	0.51***	0.12***	0.12***	0.12***	0.12***
Regulated industry dummy	-1.46	-4.27	7.09	16.81	-2.28	-5.04	-30.74	-35.09
H-share dummy	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01
B-share dummy	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01
Year 2003 dummy	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01
Year 2004 dummy	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01
Year 2005 dummy	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01
Year 2006 dummy	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01
Year 2007 dummy	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01
Wald test for state-ownership	3.10***	3.10***	3.10***	3.10***	0.03	0.30	3.84***	0.03
	Adjusted R ² =0.05				Adjusted R ² =0.05			

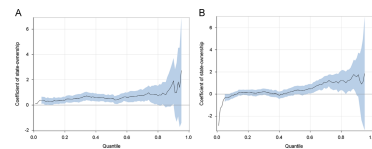


Fig. 1. Coefficients of state-ownership in quantile regressions. This figure depicts the estimates of the coefficients of state-ownership in the quantile regressions of changes in firm operating revenue and operating profit for the full sample of 1032 firms that completed the light-share structure before during 2003–2007. The quantile regression method follows that in Gozgor and Isenhardt (2015). The solid line represents coefficient values within a 0–100% quantile range. The gray area represents 95% confidence interval. **Graph A: Operating revenue; Graph B: Operating profit.**

Table 7
Post-reform stock returns and state-ownership.

The table reports regression results of the Fama-French three-factor model adjusted stock return on state-ownership for the full sample of 1032 firms that completed the light-share structure before during 2003–2007. We follow Gozgor and Isenhardt (2015) to apply quantile regression. The regression results for the 25th, 50th (median), and 75th quantiles are reported. OLS regression results are reported for reference. Data are estimated at the 1% level for OLS regressions. t-statistics are reported in parentheses. Significance: *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	25th quantile	50th quantile	75th quantile	OLS
Intercept	136.40***	131.14***	333.63***	286.14***
State-ownership (S)	0.45**	0.53***	0.82**	0.51
Non-tradable-to-tradable ratio	0.75	0.75	0.51	1.15
Log of market cap.	0.46***	0.51***	0.51***	0.51***
Change in operating revenue	0.45**	0.53***	0.82**	0.51
Change in operating profit	0.18	0.37	0.74	0.27
Log of market cap.	0.46***	0.51***	0.51***	0.51***
Regulated industry dummy	-1.46	-4.27	7.09	16.81
H-share dummy	0.03	0.03	0.02	0.01
B-share dummy	0.03	0.03	0.02	0.01
Year 2003 dummy	0.03	0.03	0.02	0.01
Year 2004 dummy	0.03	0.03	0.02	0.01
Year 2005 dummy	0.03	0.03	0.02	0.01
Year 2006 dummy	0.03	0.03	0.02	0.01
Year 2007 dummy	0.03	0.03	0.02	0.01
Wald test for state-ownership	3.10***	3.10***	3.10***	3.10***
	Adjusted R ² =0.016			

6.7 Table 7: H1 in stock returns

Dependent variable. Post-reform cumulative Fama–French-adjusted stock return.

Main finding. State-ownership is positively associated with adjusted returns at lower and median quantiles (and in OLS), but not at the upper quantile. This suggests the market prices

a stronger “privatization effect” particularly for firms whose post-reform return realization is not already extremely high.

Additional controls. The non-tradable-to-tradable ratio enters negatively and significantly, consistent with the idea that larger overhang / negotiation frictions are associated with weaker risk-adjusted performance.

6.8 Table 8: H2 (government agents’ supportive activities)

Key proxies.

- **Group listing dummy:** whether the state shareholder pursues deeper restructuring/listing of the parent group.
- **Asset injection dummy:** whether assets are injected into the listed SOE.
- **Fund raising rounds:** financing opportunities following reform.

Results (median regressions).

- These supportive activities are strongly and positively correlated with **privatization-led improvements** in operating revenue and operating profit (both unadjusted and benchmark-adjusted versions).
- Stock return regressions are noisier and less robust, which is plausible given crisis-era volatility.

Interpretation. H2 is supported: when government agents take concrete supportive actions (or enable financing), SOEs improve more. This is consistent with incentives being activated by the possibility of eventual sale/valuation of state shares.

Table 8
Post-reform changes in SOE performance and government agents’ supports.

This table reports the results of the 50th quantile (median) regressions of changes in post-reform SOE operating revenue, operating profit, and stock returns on government agents’ supportive activities. The sample consists of 633 SOEs that completed the Split-share Structure Reform during 2005–2007. The unadjusted changes in SOE operating revenue and operating profit are calculated as the differences between the Consumer Price Index-adjusted operating revenues and profits three years before and after the reform, normalized by pre-reform values. Size-industry and size-market-to-book ratio adjusted changes are calculated using the unadjusted changes in SOE variables minus median changes in the variables of the matching non-SOE benchmark portfolios by size-industry or size-market-to-book ratio. The unadjusted SOE stock returns are the Fama-French three-factor model adjusted cumulative stock returns three years after the reform. Size-industry and size-market-to-book ratio adjusted SOE stock returns are calculated using the unadjusted SOE stock returns minus the median stock returns of the matching non-SOE benchmark portfolios by size-industry or size-market-to-book ratio. The supportive activities proxies are individually included in the regressions. *t*-Statistics are reported in parentheses. Superscripts *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	Unadjusted (X)	Size-industry adjusted (X)	Size-M/B adjusted (X)
Panel A: Change in operating revenue			
Group listing dummy	137.23** (2.3)	168.50*** (3.7)	144.26** (2.3)
Asset injection dummy	5.38** (3.8)	13.32*** (3.8)	5.35** (3.4)
Wald test for group listing dummy	120.06*** (15.26)	113.81*** (14.29)	112.47*** (14.53)
Asset injection dummy	15.26*** (3.8)	14.29*** (4.0)	11.58*** (5.7)
Wald test for asset injection dummy	18.82*** (3.8)	21.61*** (4.0)	23.66*** (5.7)
Fund raising rounds	14.54*** (3.8)	15.76*** (4.0)	32.30*** (5.7)
Wald test for fund raising rounds	14.54*** (3.8)	15.76*** (4.0)	32.30*** (5.7)
Panel B: Change in operating profit			
Group listing dummy	116.23** (2.0)	96.06 (1.6)	109.94* (1.9)
Asset injection dummy	4.10** (3.4)	2.48 (3.3)	3.62* (3.4)
Wald test for group listing dummy	108.10*** (14.7)	104.27*** (10.72)	107.14*** (11.47)
Asset injection dummy	21.8*** (3.4)	10.72*** (3.3)	11.47*** (3.4)
Wald test for asset injection dummy	26.67*** (5.4)	26.04*** (5.8)	30.29*** (7.6)
Fund raising rounds	28.63*** (5.4)	33.67*** (5.8)	58.32*** (7.6)
Wald test for fund raising rounds	28.63*** (5.4)	33.67*** (5.8)	58.32*** (7.6)
Panel C: Post-reform stock returns			
Group listing dummy	15.07 (0.6)	22.81 (0.9)	14.78 (0.6)
Asset injection dummy	0.3 (0.0)	0.83 (0.1)	0.40 (0.6)
Wald test for group listing dummy	0.3 (0.0)	0.83 (0.1)	0.40 (0.6)
Asset injection dummy	0.00 (0.0)	0.01 (0.1)	0.31 (0.6)
Wald test for asset injection dummy	0.00 (0.0)	0.01 (0.1)	0.31 (0.6)
Fund raising rounds	9.01** (2.2)	6.11 (1.5)	–1.61 (0.3)
Wald test for rounds of fund raising	4.99** (2.2)	2.25 (1.5)	0.08 (0.3)

Table 9
Post-reform sales of state-owned shares and SOE performance.

This table reports the statistics of post-reform sales of state-owned shares as of October 2011. The sample consists of 633 SOEs that completed the Split-share Structure Reform during 2005–2007. Panel A reports the number of firms, number of state-owned shares sold, ratio of state-owned shares sold to total shares outstanding, and percentage of state-shareholders involved in the sales. We apply two-tail *t*-test to examine the statistical significance of these variables. *t*-Statistics are reported in parentheses except for column 2. Panel B reports the Pearson correlations between post-reform sales of state-owned shares and post-reform change in SOE operating revenue. Superscripts *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

Panel A: Post-reform sales of state-owned shares						
Industry	Obs.	Number (X in parentheses) of SOEs sold state-owned shares	Avg. number of shares sold per SOE (in millions)	Avg. % of state-owned shares sold to owned per SOE	Avg. % of state-owned shares sold to outstanding per SOE	Avg. % of state-shareholders involved in the sales
Full sample	633	160 (25.3%)	5.13*** (9.6)	2.95*** (5.8)	0.44*** (11.0)	31*** (12.7)
Utilities	69	10 (14.5%)	3.69*** (2.7)	2.66** (1.7)	0.25*** (2.5)	20*** (3.0)
Real estate	31	8 (25.8%)	4.72** (2.4)	1.24** (2.4)	0.59** (2.1)	26*** (3.2)
Comprehensive	87	25 (28.7%)	4.87*** (4.7)	2.33*** (3.2)	0.47*** (3.4)	38*** (4.9)
Manufacturing	394	108 (27.4%)	5.94*** (7.7)	3.01*** (5.0)	0.46*** (9.3)	33*** (10.6)
Commercial	52	9 (17.3%)	1.53*** (2.6)	4.91 (1.4)	0.34** (2.3)	19** (3.1)
Panel B: Correlations between sales of state-owned shares and changes in SOE operating revenue						
	Number of sales of state-owned shares	Number of state-owned shares sold (in millions)	% Of state-owned shares sold to owned	% Of state-owned shares sold to outstanding	% Of state-shareholders involved in the sales	
Unadjusted	–0.08**	–0.08**	–0.02	–0.07*	–0.09**	
Size-industry adjusted	–0.10***	–0.11***	0.00	–0.09**	–0.11***	
Size-M/B ratio adjusted	–0.09**	–0.11***	0.00	–0.08*	–0.10***	

6.9 Table 9: Post-reform sales of state-owned shares as a punitive mechanism

Panel A: Actual post-reform state-share sales exist but are small in aggregate (only a fraction of SOEs sell, and average amounts are modest relative to total shares).

Panel B: Measures of post-reform sales are **negatively correlated** with privatization-led improvements to SOE performance.

Interpretation. This is consistent with a disciplining story: the government tends to divest more in underperforming SOEs, diluting control and punishing agents who failed to raise state-share value.

6.10 Table 10 and Fig. 2: H3 (market mechanism, information discovery)

Core regression objects. Consideration and approval rate (both negotiated/realized during the reform plan vote). **Table 10 (median regressions).**

Table 10

Post-reform changes in SOE performance and the market mechanism.

This table reports the results of the 50th quantile (median) regressions of changes in post-reform SOE operating revenue, operating profit, and stock return on consideration and public investor reform plan approval rate. The sample consists of 633 SOEs that completed the Split-share Structure Reform during 2005–2007. The unadjusted changes in SOE operating revenue and operating profit are calculated as the differences between the CF-adjusted revenues and profits three years before and after the reform, normalized by pre-reform values. Size-industry and size-market-to-book ratio adjusted changes are calculated using the unadjusted changes in SOE variables minus median changes in the variables of the matching non-SOE benchmark portfolios by size-industry or size-market-to-book ratio. The unadjusted SOE stock returns are the Fama-French three-factor model-adjusted cumulative stock returns three years after the reform. Size-industry and size-market-to-book ratio adjusted SOE stock returns are calculated using the unadjusted SOE stock returns minus the median stock returns of the matching non-SOE benchmark portfolios by size-industry or size-market-to-book ratio. t-Statistics are reported in parentheses. Superscripts *, **, and *** denote the significance levels of 10%, 5%, and 1%, respectively.

	Change in operating revenue			Change in operating profit			Post-reform stock return		
	Unadjusted (t)	Size-ind. adj. (t)	Size-M/B adj. (t)	Unadjusted (t)	Size-ind. adj. (t)	Size-M/B adj. (t)	Unadjusted (t)	Size-ind. adj. (t)	Size-M/B adj. (t)
Consideration	-1.06*** (3.1)	-0.84*** (2.5)	-1.16*** (3.2)	-0.81* (1.9)	-0.72* (1.7)	-1.10*** (2.4)	0.63 (1.1)	0.72 (1.6)	0.64 (1.6)
Approval rate (%)	1.45* (1.8)	1.73** (2.0)	1.97** (2.3)	2.02** (2.0)	1.61* (1.7)	2.42*** (2.6)	0.04 (0.1)	-0.07 (0.1)	2.46*** (2.2)
State-ownership (%)	0.17 (0.6)	0.35 (1.2)	0.03 (0.1)	-0.22 (0.7)	0.08 (0.2)	-0.36 (1.2)	0.00 (0.0)	0.09 (0.2)	0.11 (0.3)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wald test for consideration	9.79***	6.10***	10.52***	3.57*	2.82*	5.97***	1.19	0.05	2.44
Wald test for approval rate	3.15*	3.91**	5.14**	3.81**	2.83*	6.57***	0.13	0.11	4.75**

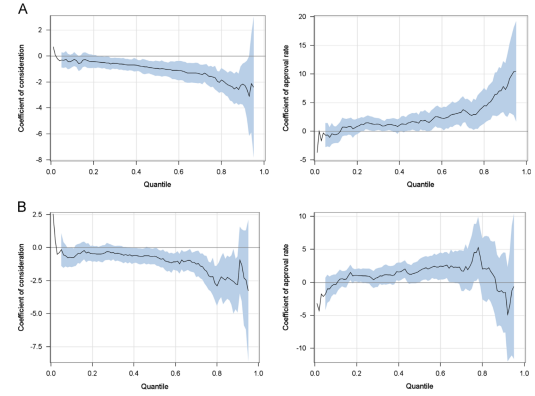


Fig. 2. Coefficients of consideration and reform plan approval rate in quantile regressions for privatization-led improvements to SOE performance. This figure depicts the estimates of the coefficients of consideration and public investor reform plan approval rate in quantile regressions. The dependent variables are changes in the size-industry adjusted SOE operating revenue and operating profit. The quantile regression method follows that in Koenker and Gelfand (1979). The solid line represents coefficient values within a 0–100% quantile range. The gray area represents 95% confidence interval. [Graph A: Size-industry adjusted operating revenue](#), [Graph B: Size-industry adjusted operating profit](#).

- **Consideration:** negatively associated with subsequent improvements in operating revenue and operating profit (robust across unadjusted and benchmark-adjusted outcomes).
- **Approval rate:** positively associated with subsequent improvements in revenue and profit, and in some specifications with stock returns.

Fig. 2 (quantile robustness). The consideration coefficient stays negative and the approval coefficient stays positive across quantiles for size-industry adjusted revenue/profit improvements.

Interpretation. The negotiation process appears to embed information: investors demand more up-front compensation (higher consideration) when expected future performance improvements are weaker, and they approve plans more when stronger improvements are expected. This supports the paper’s emphasis on *market mechanism* as an information discovery and incentive-communication device.

7 Conclusion

7.1 Summary

The Split-Share Structure Reform (SSSR, 2005–2007) converted previously non-tradable state shares in listed firms into tradable shares (after lockups and negotiated compensation). The authors interpret this reform as opening the gate to *secondary privatization*: even without immediate

transfers of control, the *expectation* that state shares can eventually be monetized in the secondary market changes incentives for government agents and incumbent managers.

The paper’s main conclusion is that the reform produced a **fast and economically large real response in SOEs**: SOEs expanded **output, profits, and employment** after the reform, but this expansion **did not come from a deep improvement in operating efficiency or corporate governance**. Instead, the evidence is most consistent with an **incentive channel**: the expectation of future privatization (and the associated importance of state-share value) induced government agents to take *quick performance-enhancing actions* (e.g., supportive restructuring, asset injections, and financing opportunities).

Methodologically, the paper emphasizes that the SSSR used a **market mechanism** (negotiation, consideration, and public-investor voting) that served as an **information discovery and alignment device** between the government (as controlling shareholder) and public investors.

7.2 Key empirical conclusions

1. **SOE output rises sharply after the reform (and more than non-SOEs).** The paper reports that operating revenue increases significantly post reform. For the full sample, the median increase is about **73%**. The median increases for **non-SOEs** and **SOEs** are about **57%** and **84%**, respectively. The gradient across state-ownership intensity is monotone: the median revenue increase rises from roughly **47%** (pure private group) to about **92%** (highest state-ownership group).
2. **Profits and employment increase, but operating efficiency is not the central margin.** Operating profit rises with a median increase around **45%** for the full sample (about **44%** for non-SOEs and **50%** for SOEs), while employment increases with a median change around **13%**. Crucially, the paper’s overall reading is that SOEs expand *without sacrificing* operating efficiency, rather than showing a clear governance-driven efficiency revolution.
3. **Corporate governance does not “fundamentally” improve absent ownership change.** Even though some transaction-level proxies can move, the paper concludes there is no strong evidence that the reform improved SOE corporate governance in a deep way, because the reform did not fundamentally reallocate control rights or ownership structure.
4. **The mechanism is incentives + supportive actions by government agents.** Improvements in SOE performance are positively correlated with measures that proxy for government agents’ privatization-related incentives (i.e., incentives to increase the value of state-owned shares) and with observable supportive activities such as group restructuring/listing, asset injections, and fundraising opportunities.
5. **The market mechanism matters: consideration and approval rates behave like informative prices.** The negotiated reform terms embed beliefs about future outcomes: higher expected performance is associated with *lower* required consideration to tradable shareholders and *higher* approval rates. This supports the view that the SSSR’s market-based process helps align government and public investors through information discovery.

7.3 Economic intuition: why these results fit together

A coherent way to understand the paper is through an incentive-and-constraints logic:

- **Why performance rises quickly:** once state shares become legitimately tradable, the government (and its agents) becomes more directly exposed to market valuation—because higher prices increase the option value of future divestment and reduce the political/financial cost of privatization. This generates an immediate incentive to improve observable performance (output/profits/employment) through actions that can be implemented quickly.
- **Why governance does not transform:** without a major change in controlling rights or monitoring institutions, the deeper agency problems of SOEs (soft budget constraints, political objectives, weak boards, tunneling opportunities) do not disappear. Hence, the reform can produce sizeable real expansion while leaving governance/efficiency largely unchanged.
- **Why the “market mechanism” is central:** the reform requires negotiation and investor approval. Consideration and approval rates therefore serve as equilibrium objects that reflect bargaining power and information about expected outcomes. This mechanism is more effective than purely administrative privatization orders because it creates an explicit channel for aligning interests and revealing information when privatization becomes politically and economically delicate.

7.4 Takeaway

By making state shares tradable, China’s Split-Share Structure Reform created a secondary-privatization incentive that rapidly boosted SOE output, profits, and employment, while leaving governance and efficiency largely unchanged; the reform’s negotiated market mechanism helped align the government and public investors through information discovery.